

COMPTE RENDU DU TP 1.1 DE CYBERSECURITE

CHAPITRE 1 : RECONNAISSANCE & THREAT INTELLIGENCE

PARTIE 1- découverte des machines actives

- ✓ Kali linux (192.168.56.103)
- Machine cible :
- ✓ Windows 7 :192.168.56.104
- ✓ Ubuntu server : 192.168.56.101

```
(fotsing@angela)-[~]
$ ping 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data.
64 bytes from 192.168.56.104: icmp_seq=1 ttl=128 time=1.87 ms
64 bytes from 192.168.56.104: icmp_seq=2 ttl=128 time=1.39 ms
64 bytes from 192.168.56.104: icmp_seq=3 ttl=128 time=1.17 ms
64 bytes from 192.168.56.104: icmp_seq=4 ttl=128 time=1.57 ms
64 bytes from 192.168.56.104: icmp_seq=5 ttl=128 time=1.55 ms
64 bytes from 192.168.56.104: icmp_seq=6 ttl=128 time=1.66 ms
^C
— 192.168.56.104 ping statistics —
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 1.174/1.537/1.874/0.218 ms

(fotsing@angela)-[~]
$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.61 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.56 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.10 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.43 ms
^C
— 192.168.56.101 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.432/1.674/2.099/0.253 ms
```

1- PRÉSENTATION DE NMAP

- ✓ Définition et rôle du Nmap

Nmap (Network Mapper) est un outil open source de scan réseau. Il permet de découvrir des hôtes actifs, les ports ouverts, les services en écoute, les versions logicielles, les systèmes d'exploitation, et bien plus.
- Contexte d'utilisation :
 - En cyber défense : pour cartographier son propre réseau pour fermer les ports inutiles et détecter des machines non autorisées.

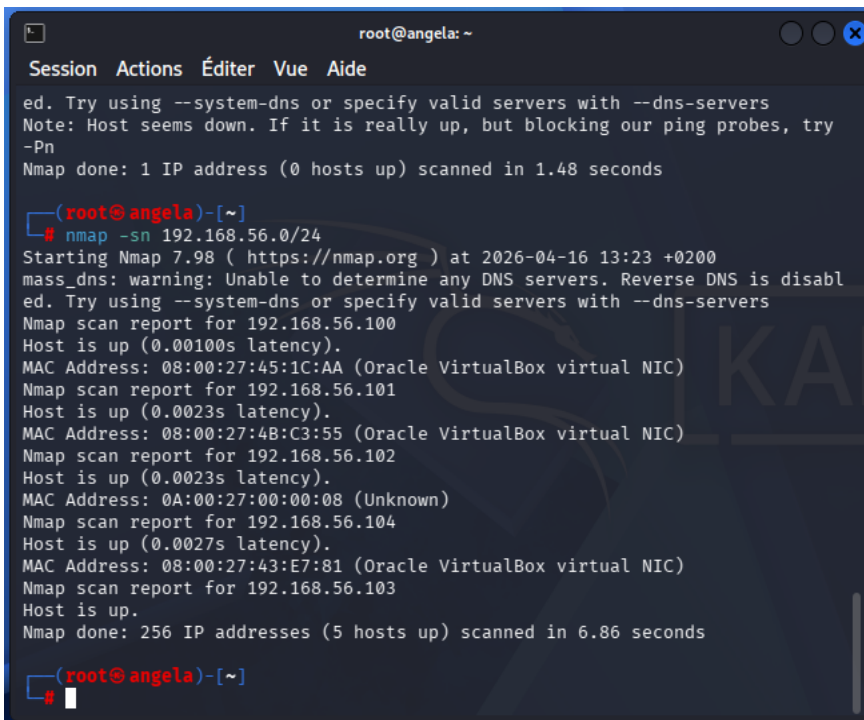
- En attaque (pentest) : phase de reconnaissance pour identifier des cibles potentielles et leur vulnérabilités.
- Il est utilisé dans les audits de sécurité, les tests d'intrusion et le diagnostic réseau.

2- ANALYSE DES COMMANDES UTILISÉES

a- Option -sn

Résultat de la commande `nmap -sn 192.168.56.0/24`

L'image suivante montre la capture de la commande.



```

root@angela: ~
Session Actions Éditer Vue Aide
ed. Try using --system-dns or specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, try
-Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.48 seconds

(root@angela)-[~]
# nmap -sn 192.168.56.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 13:23 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl
ed. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.100
Host is up (0.00100s latency).
MAC Address: 08:00:27:45:1C:AA (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up (0.0023s latency).
MAC Address: 08:00:27:4B:C3:55 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up (0.0023s latency).
MAC Address: 0A:00:27:00:00:08 (Unknown)
Nmap scan report for 192.168.56.104
Host is up (0.0027s latency).
MAC Address: 08:00:27:43:E7:81 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.103
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 6.86 seconds

(root@angela)-[~]
#

```

Permet de détecter les machines actives sans scanner les ports et utilisé pour un scan rapide du réseau.

b- Option -sS

Scan SYN (semi-ouvert), plus discret que les autres scans.

Permet de contourner certains systèmes de détection. Résultat de la commande dans l'image ci-dessus.

```
fotsing@angela: ~  
Session Actions Éditer Vue Aide  
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=1.85 ms  
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=1.59 ms  
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=1.67 ms  
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=1.62 ms  
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=2.16 ms  
64 bytes from 192.168.56.101: icmp_seq=21 ttl=64 time=1.53 ms  
64 bytes from 192.168.56.101: icmp_seq=22 ttl=64 time=1.65 ms  
^C  
— 192.168.56.101 ping statistics —  
22 packets transmitted, 22 received, 0% packet loss, time 21030ms  
rtt min/avg/max/mdev = 1.393/1.799/3.219/0.380 ms  
  
(fotsing@angela)-[~]  
$ nmap -sS 192.168.56.101  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 17:34 +0200  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.101  
Host is up (0.0015s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
MAC Address: 08:00:27:4B:C3:55 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.63 seconds  
  
(fotsing@angela)-[~]  
$
```

c- Option -sV

Permet d'identifier les versions des services sur les ports ouverts.

Très utile pour détecter des vulnérabilités. Résultat de la commande dans l'image ci-dessus.

```
fotsing@angela: ~  
Session Actions Éditer Vue Aide  
Host is up (0.0015s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
MAC Address: 08:00:27:4B:C3:55 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.63 seconds  
  
(fotsing@angela)-[~]  
$ nmap -sV 192.168.56.101  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 17:36 +0200  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.101  
Host is up (0.0021s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)  
MAC Address: 08:00:27:4B:C3:55 (Oracle VirtualBox virtual NIC)  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 3.30 seconds  
  
(fotsing@angela)-[~]  
$
```

e- Option -O

Permet de détecter le système d'exploitation de la machine cible.

Résultat de la commande dans l'image ci-dessus nous un système linux et Windows

```
fotsing@angela: ~  
Session Actions Éditer Vue Aide  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 3.30 seconds  
  
(fotsing@angela)-[~]  
$ sudo nmap -O 192.168.56.101  
[sudo] Mot de passe de fotsing :  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 17:39 +0200  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.101  
Host is up (0.0015s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
MAC Address: 08:00:27:4B:C3:55 (Oracle VirtualBox virtual NIC)  
Device type: general purpose  
Running: Linux 4.X|5.X  
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5  
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)  
Network Distance: 1 hop  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 3.60 seconds  
  
(fotsing@angela)-[~]  
$
```

f- Option -A

Scan complet incluant détection OS, services, scripts et traceroute. . Résultat de la commande dans l'image ci-dessus.

```
fotsing@angela: ~  
Session Actions Éditer Vue Aide  
Nmap scan report for 192.168.56.101  
Host is up (0.0018s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)  
| ssh-hostkey:  
| 3072 f7:26:6f:34:6c:34:e6:b9:2b:cd:88:90:c8:62:81:d5 (RSA)  
| 256 d6:66:bf:7d:a8:47:35:b6:31:cb:67:74:b8:ba:7b:42 (ECDSA)  
|_ 256 17:b5:e6:36:8f:f2:1b:a1:df:51:2d:e5:d2:ab:92:f0 (ED25519)  
MAC Address: 08:00:27:4B:C3:55 (Oracle VirtualBox virtual NIC)  
Device type: general purpose  
Running: Linux 4.X|5.X  
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5  
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)  
Network Distance: 1 hop  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
TRACEROUTE  
HOP RTT      ADDRESS  
1 1.83 ms 192.168.56.101  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 6.31 seconds  
  
(fotsing@angela)-[~]  
$
```

3- RÉSULTATS OBTENUS

Les machines détectées incluent les ordinateurs (win10 et Ubuntu-server 20)

Les ports ouverts courants sont :

- Pour Windows 10

ports	statut	services
135/tcp	open	msrpc
139/tcp	open	Netbios-ssn
445/tcp	open	Microsoft-ds(SMB)

- Pour Ubuntu-server

22/tcp	open	ssh
--------	------	-----

Les services identifiés permettent de comprendre les applications actives sur les machines.

4- ANALYSE DE SÉCURITÉ

Certains services comme SMB et SSH sont considérés comme dangereux car ils ne sont pas sécurisés.

Les ports inutilisés doivent être fermés pour réduire la surface d'attaque.

Les risques incluent l'accès non autorisé, le vol de données et les attaques informatiques.

5- UTILISATION AVANCÉE DE NMAP

a- Option -p-

Permet de scanner tous les ports (1 à 65535). Dans les images suivantes on peut voir les ports ouverts présents dans les deux machines ainsi que leur service.


```
(fotsing@angela)-[~]
$ nmap -p- 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 20:41 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.0015s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:4B:C3:55 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 113.49 seconds

(fotsing@angela)-[~]
$ nmap -p- 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 20:50 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.104
Host is up (0.0016s latency).
Not shown: 65526 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:43:E7:81 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 133.18 seconds

(fotsing@angela)-[~]
```

b- Option --script vuln

Permet de détecter automatiquement les vulnérabilités connues. Image de la commande ci-dessus nous montre que le service de partage de fichier est accessible mais vulnérable sur la machine Windows, car le système bloque les tentatives de connexion.

```
(fotsing@angela)-[~]
$ nmap --script vuln 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 20:55 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.104
Host is up (0.0013s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 08:00:27:43:E7:81 (Oracle VirtualBox virtual NIC)

Host script results:
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_smb-vuln-ms17-010:
|  VULNERABLE:
|    Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|    State: VULNERABLE
|    IDs: CVE:CVE-2017-0143
|    Risk factor: HIGH
|    A critical remote code execution vulnerability exists in Microsoft SMBv1
|      servers (ms17-010).
|
|    Disclosure date: 2017-03-14
|    References:
|      https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
```

Sur la machine Ubuntu aucune vulnérabilité malgré le port ssh soit ouvert l'image suivante nous montre le résultat.

```
(fotsing@angela)-[~]
$ nmap --script vuln 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 21:05 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.101
Host is up (0.0018s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:4B:C3:55 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 3.72 seconds

(fotsing@angela)-[~]
$
```

c- Option -T4

Permet d'accélérer la vitesse du scan. A T4 la vitesse augmente on peut passer de 15s à 6s. L'image suivante nous montre le résultat.

```
(fotsing@angela)-[~]
$ nmap -T4 -A 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 21:07 +0200
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.104
Host is up (0.0014s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows 7 Home Basic 7601 Service Pack 1 microsoft-ds (workgroup: WORKGROUP)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49156/tcp  open  msrpc        Microsoft Windows RPC
49157/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:43:E7:81 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 2008|7|Vista|8.1
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_vista cpe:/o:microsoft:windows_8.1
OS details: Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Windows 8.1
Network Distance: 1 hop
Service Info: Host: WIN7; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb-os-discovery:
|_ OS: Windows 7 Home Basic 7601 Service Pack 1 (Windows 7 Home Basic 6.1)

Active Windows
```

d- Option -Sv -Sc -O

Scanne avec détection d'os et version des protocoles.

CONCLUSION

Nmap est un outil puissant pour analyser et sécuriser un réseau. Son utilisation doit être légale et éthique. C'est aussi un outil open source de scan réseau. Il permet de découvrir des hôtes actifs, les ports ouverts, les services en écoute, les versions logicielles, les systèmes d'exploitation, et bien plus.

